

AP CSP | Lesson 23: Systems & Networks

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B includes the cooperating devices and programs. A is too narrow; C lacks the shared working system; D wrongly requires Internet access.

Q2. Answer: C

C is the connectivity requirement. A is not a path rule; B confuses validity with optimization; D uses drawing position instead of links.

Q3. Answer: A

A describes the lookup. B confuses lookup with content transfer; C confuses addressing with routing; D is unrelated to this function.

Q4. Answer: D

D distinguishes service and network. A equates part with whole; B confuses a service with local access technology; C ignores its reliance on protocols.

Q5. Answer: A and C

A supports interoperability; C gives the capacity unit. B invents a requirement; D infers capacity from topology alone. Select A and C only.

Q6. Answer: B

B: $90/30 = 3$ seconds. A reverses the ratio; C subtracts unlike quantities; D multiplies data by rate.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: Sensor, server or display controller; report readings; connected devices exchange data; sensor readings enter the server.

1 mark: Identifies a device capable of running a program.

1 mark: Gives a purpose involving their coordinated work.

1 mark: Explains connected devices can exchange data.

1 mark: Identifies readings as the server input.

Q8. 4 marks

Answer: Example S-A-C-T: 3 hops. S-B-C-T lacks B-C. Graph positions do not give a routing metric.

1 mark: Gives a path using only stated edges.

1 mark: Correctly counts links in that path.

1 mark: Identifies missing B-C.

1 mark: Distinguishes drawing position from a stated routing rule.

Q9. 4 marks

Answer: DNS returns address information; the address directs communication toward a destination; the response contains requested content; lookup and content delivery are separate functions.

1 mark: Gives address information as the lookup result.

1 mark: Explains destination addressing.

1 mark: Identifies page/content data in the response.

1 mark: Distinguishes name resolution from content serving.

Q10. 4 marks

Answer: 4 seconds; bits per second; $80/40 = 2$ seconds; no, connectivity alone does not establish link capacity or use.

1 mark: Calculates 4 seconds.

1 mark: Gives bits per second or an appropriate scaled form.

1 mark: Calculates 2 seconds at 40 megabits/s.

1 mark: Explains that extra routes do not alone specify capacity.

Q11. 4 marks

Answer: Internet: interconnected networks using open protocols. Web: linked pages, programs and files. HTTP. Other services or paths may still work.

1 mark: Defines the Internet as interconnected networks with shared protocols.

1 mark: Describes the Web as linked pages/programs/files.

1 mark: Names HTTP.

1 mark: Explains that failure of one service is not failure of all connectivity.

Q12. 4 marks

Answer: Agreed behavior rules; creators can implement the same rules; a connection to an Internet-connected device/network; finite link capacity or other resource limits remain.

1 mark: Defines agreed rules governing communication behavior.

1 mark: Explains compatible implementations using shared open rules.

1 mark: Identifies a path to an Internet-connected device/network.

1 mark: Identifies a resource limit unaffected by mere protocol agreement.